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Scheduling stochastic distributed flexible job shops using an multi-objective evolutionary algorithm with simulation evaluation

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ABSTRACT

The trend of reverse globalisation prompts manufacturing enterprises to adopt distributed structures with multiple factories for improving production efficiency, meeting customer requirements, and responding disturbance events. This study focuses on scheduling a distributed flexible job shop with random job processing time to achieve minimal makespan and minimal total tardiness. First, a stochastic programming model is established to formulate the concerned problems. Second, in accordance with the natures of two objectives and randomness, an evolutionary algorithm incorporating an evaluation method is designed. In it, population-based and external archive-based search processes are developed for searching candidate solutions, and the evaluation method integrates stochastic simulation and discrete event simulation to calculate objective values of acquired solutions. Finally, a mathematical optimisation solver, CPLEX, is employed to validate the developed model and optimisation approach. A set of cases is solved to verify the performance of the proposed method. The comparisons and discussions show the superiority of the proposed method for handling the problems under study.

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1. Introduction

Recently, the world situation greatly changes, which profoundly affects the global supply chains. In the globalisation era, managers prefer to take the lean operation as a main criterion, where much effort is made to choose collaborative members such as suppliers, manufactures and carriers having the lowest cost to enhance the overall profit (Becerra, Mula, and Sanchis 2023). However, the recent years witness the trend of reverse globalisation as a series of global events occur. For instance, the global epidemic of COVID-19 results in the shutdown of factories and increase of transportation cost, etc. Besides, the Russia-Ukraine war unexpectedly breaks out, along with the wide international sanctions and energy price rise. All the global events certainly influence the organisation of supply chains, causing huge uncertainties of manufacturing industries. In the context of such situations, it is very important to provide required products for customers rather than excessively emphasise the difference of manufacturing cost among diverse regions. Hence, the

global markets will be further segmented (Xu et al. 2022). Unlike the lean supply chains with minimal operation cost, the distributed structures with redundant resource configuration might cope with the reverse globalisation and uncertain events (Badakhshan and Ball 2023; Shi and Mena 2023; Wu et al. 2023). In recent years, manufacturing enterprises gradually establish multiple factories or production shops in different geographical locations and assign production tasks to them for reducing the risk incurred by the disturbance and attack of uncertain events (Choi et al. 2022).

To this end, a burgeoning distributed manufacturing is taking the place of the single-factory production process. Meanwhile, the wide use of advanced techniques, e.g. cyber-physical systems and Artificial Intelligence, promotes the implementation of distributed manufacturing (Siqin et al. 2022). Recently, many enterprises apply the paradigm of distributed manufacturing to their production processes such as automotive and steel-making industries (Fu et al. 2021a). In such circumstances,

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the concrete information of factories, such as resource deployment, machine function and production environment, must be known in advance. Meanwhile, the job processing routes and the needed machines should be determined beforehand. Significantly, all the involved factories must share production information to operate in a collaborative manner.

Production scheduling has vital effects on the management operations of manufacturing systems (Fu et al. 2023a; Hou et al. 2023; Neumann et al. 2023). Currently, scheduling distributed manufacturing systems, i.e. distributed scheduling, has captured an amount of concern (Li et al. 2022a; Pan et al. 2023). However, addressing distributed scheduling problems still have great challenges for academia and industry (Lei and Su 2023; Pan, Lei, and Wang 2022; Shao, Shao, and Pi 2021; Wang et al. 2023; Zhao, Ma, and Wang 2022). Job shop scheduling problems (JSPs) have emerged as a key issue in manufacturing and service systems (Carlucci, Renna, and Materi 2023). By extending JSPs with multi-purpose machines, researchers further develop flexible JSPs (FJSPs), in which each operation is permitted to be fabricated by any one of candidate machines. Hence, FJSPs need to determine both machine assignment and operation sequencing (Thenarasu et al. 2023).

Distributed FJSPs (DFJSPs) have extensive applications in various industrial areas (Meng et al. 2020). This research is motivated by a real construction equipment manufacturing system, which can be modelled as a DFJSP. In practice, construction equipment manufacturing enterprises need to produce various machines such as tube mill, rotary kiln, roller mill, stacker-reclaimer and roll squeezer. They are comprised of diverse subassemblies and each of them needs to be performed via a series of operations on drilling, grinding, milling, threading and broaching machines, etc. As usual, customers require personalised customisation on the subassemblies. They thus have multiple processing routes, where each operation can be done on any one of candidate machines. At present, enterprises are developing multiple factories or shops to improve production efficiency. In reality, many factors, such as cutter wear abrasion and worker proficiency, affect the processing time of subassemblies on machines. Hence, it is intractable to precisely know subassemblies' processing time beforehand. Generally, managers need to concern two criteria: production cost and customer satisfaction, formulated as reaching minimal makespan and total tardiness. To sum up, the above-mentioned industry scenario can be abstracted as a DFJSP with minimising makespan and total tardiness in uncertain environments. The natures of multiple optimisation objectives and uncertainties result in the difficulties of making well-informed and highly reliable

scheduling decisions for this problem. Inspired by the above production scenario, this work first time studies the modelling and optimisation methods of the DFJSP with multiple optimisation objectives and uncertainties.

Section 2 sums up the relevant studies and summarises the main contributions. Section 3 formulates the mathematical model and analyzes the difficulties of solving it. Section 4 provides the procedure of solution algorithms. Section 5 makes comparative experiments and dissects acquired results. Finally, Section 6 concludes this paper and prospects new topics.

2. Literature review

2.1. Related studies

DFJSPs have received much attention in recent years. Jeong and Yim (2009) analysed the decision-making process of a distributed JSP and proposed a cooperation method using Lagrangian relaxation to achieve completion time minimisation. Şahman (2021) devised a spotted hyena method for handling the distributed JSP to achieve minimal makespan. Azab and Naderi (2014) formulated an integer programming to describe a DFJSP and developed a greedy heuristic based on iterative insertion operations. Their another work (Naderi and Azab 2015) considered minimising makespan in a DFJSP and presented a simulated annealing (SA) method. Chaouch, Driss, and Ghedira (2017) designed an enhanced ant colony optimisation (ACO) approach to achieve makespan minimisation for a DFJSP. They also proposed a dynamic assignment rule and an ACO method (Chaouch, Driss, and Ghedira 2019). Wu, Liu, and Zhao (2019) considered assembly operations in DFJSPs. They employed a differential evolution method aiming at minimising total earliness/tardiness and total cost. Meng et al. (2020) designed a constraint programming according to interval decision variables and domain filtering methods to minimise makespan for handling DFJSPs. Wang et al. (2021b) came up with a DFJSP having timely response requirement and energy efficiency. A game method was provided to cope with it. Noteworthy, researchers used many meta-heuristic approaches, e.g. genetic algorithm (GA) (Chang and Liu 2017; De Giovanni and Pezzella 2010; Lin et al. 2020; Liu, Chen, and Chou 2014; Lu et al. 2018), chemical reaction optimisation (Marzouki, Driss, and Ghédira 2018), teaching-learning-based optimisation (Tang et al. 2022), grey wolf optimisation (Li et al. 2022b), whale optimisation (Zhao et al. 2023) methods, to address DFJSPs with minimising makespan.

In addition to the afore-mentioned studies which focused on single-objective optimisation for DFJSPs, other recent studies have examined multi-objective

DFJSPs. Xie et al. (2019) sought to reach minimal makespan and minimal energy consumption in a DFJSP. An artificial bee colony method was devised to figure out it. Luo et al. (2020) worked out a DFJSP where jobs were allowed to be transferred among factories. They adopted a memetic approach to achieve makespan, maximum workload and energy consumption minimisation. Furthermore, their recent work (Luo et al. 2022) considered a DFJSP with worker arrangement to minimise makespan as well as maximum workload regarding workers and machines. A memetic algorithm was devised to address it. Du et al. (2021) devoted to realising makespan and energy consumption minimisation in settling DFJSPs with crane transportation. They designed an estimation of distribution method for dealing with it. Sang and Tan (2022) proposed a many-objective DFJSP with consideration of economic and green objectives. They designed a many-objective memetic algorithm to work out it. Xu et al. (2021) put forward a DFJSP with outsourcing operations to optimise makespan, quality and carbon emission. A GA incorporating a tabu search method was given. Jiang, Wang, and Peng (2020) considered a DFJSP to find minimal makespan and energy consumption. An evolutionary algorithm (EA) with a decomposition strategy was devised to handle it.

Different from the above work, some studies concerned uncertainties in scheduling distributed production processes. However, nearly all of them concentrated on working out distributed flow shop scheduling problems (Fu et al. 2019; Zheng, Wang, and Wang 2020), and little attention was paid on uncertain DFJSPs. Hsu, Kao, and Lai (2016) considered a distributed JSP with fuzzy constraints. They gave a fuzzy constrained negotiation to address it. Mahmoodjanloo et al. (2022) proposed a rescheduling method via machine re-configurability for a distributed JSP with consideration of uncertain events. They designed an equilibrium optimizer algorithm to minimise total weighted lateness.

2.2. Discussions

This work makes a concise summarisation regarding the relevant studies on DFJSPs in Tables A1 and A2 in the Appendix. Via analysing the prior research on distributed JSPs, we find that they share the following features:

- 1) Such problems having single- and multi-objective optimisation attract much concern. Nevertheless, nearly all consider minimising makespan and maximum workload from the view of manufacturers, less attention is paid to improving customer satisfaction to win more potential orders, such as minimising tardiness incurred by delivery delay.
- 2) Most prior studies focus on scheduling distributed job shops in deterministic environments. Nevertheless, there exist some uncertainties in an actual process such as processing time variation and machine failure. It is worth mentioning that the existing work pays no attention to multi-objective optimisation in uncertain circumstances. Well-balanced and high-reliable schedules must account for such problems.
- 3) Meta-heuristic approaches have been extensively used to figure out distributed JSPs, and corresponding results show their strong competitiveness.

2.3. Main contributions

By summing up the existing studies, we find that DFJSPs with multiple optimisation objectives and uncertainties receive little attention over the past years. Hence, this article studies the modelling and optimisation of such problems. The difference between our work and existing studies is provided in Table A2 in the Appendix. Via performing comparisons and analysis, we make the next new contributions:

- 1) This work proposes a DFJSP with uncertainties to reduce production cost and improve customer satisfaction. Therefore, we address a distributed flexible job shop model having random processing time to achieve minimal makespan and minimal total tardiness. Accordingly, we formulate a stochastic programming model to define this issue.
- 2) Unlike the previous studies on deterministic DFJSPs, in which amount of effort is devoted to searching candidate solutions in a solution domain, we must seek for superior solutions and evaluate their performance since the random features. Thus, we devise a hybridisation of an EA and an evaluation method (MOEA-EM) to cope with this problem.
- 3) By comparing the developed method with non-dominated sorting GA II (NSGA-II) (Deb et al. 2002), multi-objective EA according to decomposition (MOEA/D) (Zhang and Li 2007), bi-objective multi-start SA method (BMSA) (Lin and Ying 2013), improved memetic algorithm (IMA) (Luo et al. 2022) and mathematical programming solver, CPLEX, we verify that MOEA-EM can achieve more outstanding results in resolving the problem under study.

2.4. Managerial insights

This work investigates a construction equipment manufacturing process that is modelled as a DFJSP with cost-awareness and tardiness-aversion criteria in uncertain

environments. The results uncover the following implications:

- 1) This work offers competitive modelling, optimisation and performance assessment approaches to solve this problem. Simulation experiments and comparisons verify that the developed models and methods can provide excellent solutions. The results can assist managers and decision-makers in making much more informed and more reliable decisions for scheduling a distributed flexible job shop having two optimisation objectives and random processing time. The research findings are ready to be applied in a real-life construction equipment manufacturing process's scheduling.
- 2) To implement the designed method, the detailed information of factories in distributed manufacturing systems, e.g. resource allocation and machine flexibility, needs to be acquired. Subsequently, the processing routes of jobs (namely, customer orders) should be made, and the required machines of sub-assemblies are correspondingly determined. Moreover, the production data must be acquired and analysed in advance, and the fittest stochastic distributions should be mathematically developed. The above-mentioned information is input into the formulated method, and subsequently it offers a group of alternative decisions to managers for reference.
- 3) Under the circumstances of reverse globalisation and variable production environments, distributed manufacturing contributes to assisting enterprises in resisting uncertain disturbance such as machine breakdown and processing time variation, and even factory stagnation. The resilience of manufacturing systems is accordingly improved via using such distributed production processes. However, enhancing the resilience brings a new challenge to schedule distributed manufacturing systems since we must make effective strategies to cope with diverse disruptions incurred by disturbance and attack. Hence, it is more complex than this work's consideration, and we will explore to increase the resilience of distributed manufacturing systems based on this research.

3. Problem definition

3.1. Problem statement

The proposed DFJSP includes multiple factories for processing a group of jobs, and each one is modelled as a flexible job shop having multi-function machines. Each job involves multiple operations, and each of them ought to be fabricated on one out of the candidate machines

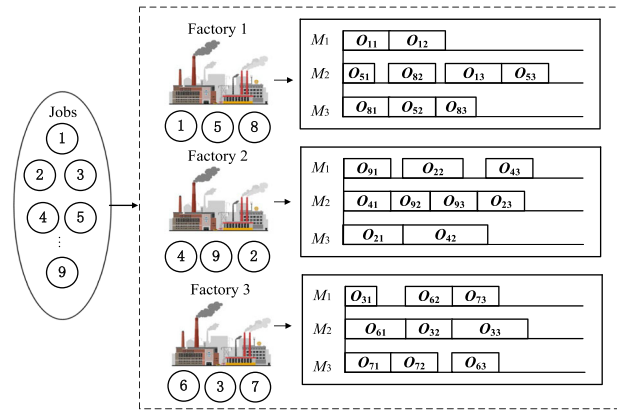


Figure 1. Illustration of the considered distributed flexible job shop.

as given sequence. Job processing time is random and follows known probabilistic distributions. All jobs have delivery requirements, namely, a tardiness occurs in case that a job's completion time is greater than a specified due date. An executable schedule should satisfy the restrictions: (1) At time 0, all machines at factories are available, and all jobs wait to be processed; (2) Each job must be produced at only one factory; (3) At a time, each machine is permitted to fabricate one job at the utmost, and similarly each job is permitted to be done by one machine at the utmost; and (4) Interruption of machines is prohibited.

To find a feasible schedule, we must make three decisions: factory assignment of jobs, machine assignment of jobs and operation sequence on machines. The objectives focus on reaching minimal makespan and minimal total tardiness. Figure 1 diagrams a solution of an instance having nine jobs, three factories and three machines per factory. Each job owns three operations, and each of which ought to be fabricated by one of three machines at a factory.

For the DFJSP under study, achieving minimal makespan and reaching minimal total tardiness are inconsistent. This work has proved this inference in Section II in the Appendix. Thus, the DFJSP having the two criteria is a multi-objective optimisation problem. Consequently, we have Property 3.1, indicating that this problem has multiple, and even infinite optimal solutions.

Property 3.1: There exists an inconsistent relationship regarding reaching minimal makespan and minimal total tardiness in the DFJSP under consideration.

Notice that the job processing time on machines follows stochastic distributions that are mutually independent and known in advance. Nevertheless, it is unable to derive the concrete distributions of job completion time since the whole production process contains machine

stand-by time and job waiting time. Thus, we deduce Property 3.2, implying that it is intractable to reach an optimal schedule.

Property 3.2: Although the stochastic distributions of job processing time on machines are mutually independent and known in advance, the concrete expression regarding their completion time is yet indeterminable.

3.2. Stochastic programming model

To formulate the DFJSP under study, this work employs the following symbols:

Notations:

- F : set of factory index, $F = \{1, 2, \dots, f\}$, in which f indicates the quantity of factories.
- N : set of job index, $N = \{1, 2, \dots, n\}$, in which n is the quantity of jobs.
- i : job index, $i \in N$.
- n_i : quantity of operations of job i .
- i_j : job index associated with operation j , $i_j \in N$.
- l : factory index, $l \in F$.
- O : operation index set, $O = \left\{1, 2, \dots, n_i, n_i + 1, \dots, \sum_{i \in N} n_i\right\}$.
- N' : index set including both factories and jobs, $N' = \left\{1, 2, \dots, f, f + 1, \dots, n_i, n_i + 1, \dots, f + \sum_{i \in N} n_i\right\}$.
- j, j' : index for N' , $j, j' \in N'$.
- M_l : index set of machines at factory l , $M_l = \{1, 2, \dots, m_l\}$, in which m_l is the quantity of machines at factory l .
- M'_{jl} : index set of candidate machines of operation j at factory l , $j \in N'$, $l \in F$.
- k : machine index.
- p_{jkl} : processing time of operation j on machine k at factory l .
- d_i : due date of job i .
- $h_{jj'}$: indication denoting the priority relationships of two operations of jobs. It equals 1 on condition that operation j must be handled in the front of j' in the premise of $i_j = i_{j'}$, and 0 otherwise.
- G : very large number.

Notice that p_{jkl} is random.

Decision variables:

- S_j : start time of operation j .
- C_i : completion time of job i .

- C_{max} : makespan, i.e. maximal completion time of all jobs.
- T_i : tardiness of job i .
- x_{jl} : equalling 1 on condition that operation j is allocated to factory l for processing, and 0 otherwise.
- $y_{jj'kl}$: equalling 1 on condition that machine k at factory l processes operation j then does j' , and 0 otherwise.

Notice that S_j , C_i , C_{max} , and T_i are random.

Via using the aforesaid notations and definition, this work formulates a stochastic programming model below.

$$\min E(C_{max}). \quad (1)$$

$$\min E\left(\sum_{i=1}^n T_i\right). \quad (2)$$

s. t.

$$\sum_{l \in F} \sum_{k \in M'_{jl} \cap M'_{jl'}} y_{jj'kl} = 1, \forall j \in O. \quad (3)$$

$$\sum_{k \in M'_{jl}} \sum_{j \in O} y_{jj'kl} = 0, \forall j' \in F, j' \neq l. \quad (4)$$

$$\sum_{j \in O} y_{ljjk} \leq 1, \forall k \in M'_{jl}, \forall l \in F. \quad (5)$$

$$\sum_{j \in N'} y_{jj'kl} = \sum_{j \in N'} x_{j'jkl}, \forall j' \in N', \forall k \in M'_{jl} \cap M'_{jl'}, \forall l \in F. \quad (6)$$

$$x_{jl} = \sum_{k \in M'_{jl} \cap M'_{jl'}} y_{jj'kl}, \forall j \in O, \forall l \in F. \quad (7)$$

$$x_{jl} = x_{j'l}, \forall j, j' \in O' \subseteq O, O' = \{j | j \in O \parallel i_j = i\}, \forall i \in N, \forall l \in F. \quad (8)$$

$$E(S_j + p_{jkl} - S_{j'}) \leq G \cdot (1 - y_{jj'kl}), \forall j, j' \in O, \forall k \in M'_{jl} \cap M'_{j'l}, \forall l \in F. \quad (9)$$

$$E(S_j + p_{jkl} - S_{j'}) \leq G \cdot (1 - h_{jj'}), \forall j, j' \in O, i_j = i_{j'}, \forall k \in M'_{jl}, \forall l \in F. \quad (10)$$

$$E(S_j + p_{jkl} - C_i) \leq 0, \forall i \in N, \forall j \in O, \forall k \in M_l, \forall l \in F, i \in O' \subseteq O, O' = \{j | j \in O \parallel i_j = i\}. \quad (11)$$

$$E(C_{max} - C_i) \geq 0, \forall i \in N. \quad (12)$$

$$E(C_i - T_i) \leq d_i, \forall i \in N. \quad (13)$$

$$S_j \geq 0, C_i \geq 0, T_i \geq 0, p_{jkl} \geq 0, \forall i \in N, \forall j \in N', \forall k \in M_l, \forall l \in F. \quad (14)$$

$$x_{jl} \in \{0, 1\}, y_{jj'kl} \in \{0, 1\}, \forall j, j' \in N', \forall k \in M_l, \forall l \in F. \quad (15)$$

where (1) and (2) mean that reaching minimal expected makespan and minimal expected tardiness are two objectives. (3) represents that an operation has only one succession operation. In the case that an operation is the last one, its succession is defined as its associated factory index. (4) means that the first operation on machines does not have a precedence operation. (5) indicates that an operation should be fabricated on at most one machine at a time at a factory. (6) gives the relationships of two operations on machines at a factory. (7) defines the relationships of two decision variables. (8) stipulates that a job's operations ought to be handled within one factory. (9) means that an operation's completion time should be equal to or smaller than the start time of its succession on machines at a factory. (10) limits the sequence relationships of two operations belonging to a job, i.e. an operation's completion time must be equal to or smaller than its succession operation. (11) and (12) respectively define the job completion time and makespan. (13) formulates the job tardiness. (14) and (15) regulate the employed decision variables.

In the distributed flexible job shops, there exist a few factories, and each of them is modelled as a flexible job shop. The prior work has proven that scheduling each of them to reach minimal makespan or minimal total tardiness is NP-hard (Tremblet, Thevenin, and Dolgui 2023). For the problem under consideration, we need to determine factory assignment of jobs in addition to machine assignment of jobs and operation sequence on machines like a common FJSP. Hence, the investigated problem is very difficult to be solved since it features with multiple optimisation objectives, stochasticity, and complex process. Thus, we infer Corollary 3.1, indicating that there does not exist a polynomial-time approach for addressing this problem.

Corollary 3.1: The DFJSP with random processing time to minimise makespan and total tardiness is NP-hard in ordinary sense.

3.3. Challenge of addressing the studied DFJSP

The considered problem has multiple optimisation objectives and uncertainties, posing great challenges for effectively handling it. The difficulties of solving this problem are analysed as:

- 1) According to the afore-mentioned analysis, we must do the following decisions: factory assignment of jobs, machine assignment of jobs and operation sequence on machines. Moreover, it has multiple optimisation objectives, and managers prefer to acquire a great many of non-dominated solutions. Thus, the solution domain is very great in

accordance with Property 3.1 and Corollary 3.1, and mathematical programming methods are difficult to address it within an acceptance time. It is necessary to use an effective search method with special strategies by combining the problem's features. Hence, we must design high-efficiency search approaches to find promising solutions from the solution space.

- 2) Due to the random nature of the considered problem, it is hard to exactly assess the searched solutions since Property 3.2. Thus, we must decrease the influence of random disturbance in assessing the quality of solutions. In addition, it is difficult to track the job movement in the considered distributed manufacturing system because of the complex production process. Hence, we should mimic the production process better. Consequently, a new approach needs to be devised to reduce the random disturbance and mimic the complex production process.

In accordance with the above analysis, we deem that an integration of multi-objective evolutionary algorithm and simulation method can be treated as a prospective approach for handling the considered problem. Thus, we develop such an MOEA-EM.

4. Proposed approach

Considerable attention has focused on meta-heuristics in solving complicated optimisation problems (Fu et al. 2023b; Huang, Pan, and Gao 2023; Ip et al. 2003; Ma et al. 2023a). EAs are seen as an excellent approach to tackle multi-objective optimisation problems on account of the remarkable ability to acquire multiple solutions at each iteration (Fu et al. 2020; Geng and Li 2023). The past few years have witnessed the extensive and successful use of multi-objective EAs (MOEAs) for handling distributed scheduling problems (Wang and Wang 2021a). To our best knowledge, almost all of them address the distributed scheduling models in deterministic environments, and their performance in dealing with uncertain distributed scheduling problems, especially the DFJSPs with uncertainties, should be continuously explored. This article devises an MOEA-EM considering the problem's natures of multiple objectives, stochasticity, and complex constraints. The developed MOEA and evaluation method interact with each other in a search process. The former is dedicated to searching favourable individuals from a solution domain, and all the generated individuals are given to the developed evaluation method. The latter uses stochastic simulation and discrete event simulation to calculate objective values of the delivered individuals. Then, the objective values are returned to the MOEA. Subsequently, MOEA chooses the better individuals to

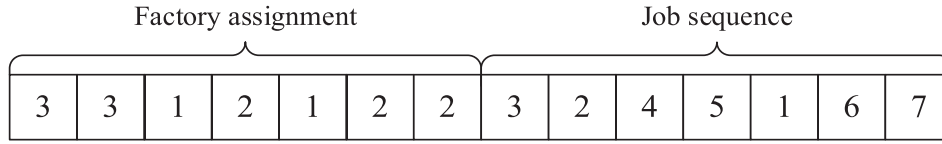


Figure 2. Illustration of solution representation methods.

construct its next population, and then enters the next iteration. The design of MOEA-EM is given below.

4.1. Solution representation

According to the above statement, we need to determine three decisions, namely factory assignment of jobs, machine assignment of jobs, and operation sequence on machines. This article adopts a double-link integer string to denote an individual (called a solution as well), i.e. $\pi = \{\pi', \pi''\}$, where $\pi' = \{\pi'_1, \pi'_2, \dots, \pi'_n\}$ is a factory assignment string and $\pi'' = \{\pi''_1, \pi''_2, \dots, \pi''_n\}$ is a job sequence string. In π' , each position ξ is associated with a job index, and π'_ξ denotes a factory index, indicating that job ξ is assigned to it, $\pi'_\xi \in [1, f]$, $\xi = 1, 2, \dots, n$. In π'' , each element π''_ξ in the ξ -th position denotes a job index, $\pi''_\xi \in [1, n]$, $\xi = 1, 2, \dots, n$, and the relative order of jobs allocated to a factory denotes their processing sequence. Figure 2 gives a solution of an example having three factories and seven jobs. From the factory assignment string, we find that factory 3 processes jobs 1 and 2, and from the job sequence string, their processing sequence is $2 \rightarrow 1$. The decisions of factory assignment and job sequence regarding other jobs can be made by the same method.

4.2. Evaluation method

The solution representation approach determines only the factory assignment and job sequence at factories, while we are unable to straightforwardly attain the decisions of machine assignment regarding jobs and operation sequence on machines at factories. Moreover, the stochasticity of job processing time results in difficulties of evaluating the objective values of solutions. Hence, this work designs an evaluation method to decode an individual into a viable schedule, and evaluate its objective values. Because the randomness and complex process of the considered distributed flexible job shop, the evaluation method improves both stochastic simulation and discrete event simulation. The former aims to assess individuals' objective values, while the latter simulates the movement of jobs at factories via using heuristic rules.

Stochastic simulation has been widely adopted to estimate probabilistic functions (Fu et al. 2020). It averages a great quantity of samples to estimate the true value. All

elements in a sample are generated from their random distributions. Suppose that π represents an assessed solution, $\bar{A}(\pi)$ and $\bar{B}(\pi)$ are severally the expected makespan and total tardiness. W denotes the total number of samples equalling to 50 in this research, $A_w(\pi)$ and $B_w(\pi)$ are respectively the makespan and total tardiness of the w -th sample. The procedure of stochastic simulation in MOEA-EM is provided as:

Step 1: Let $\bar{A}(\pi) := 0$, $\bar{B}(\pi) := 0$, $w := 1$, $W := 50$.

Step 2: Create a sample $\rho_w(\pi)$.

Step 3: Compute $A_w(\pi)$ and $B_w(\pi)$ of $\rho_w(\pi)$.

Step 4: Add $A_w(\pi)$ and $B_w(\pi)$ to $\bar{A}(\pi)$ and $\bar{B}(\pi)$.

Namely,

$$\bar{A}(\pi) := \bar{A}(\pi) + A_w(\pi), \quad \bar{B}(\pi) := \bar{B}(\pi) + B_w(\pi).$$

Step 5: Let $w := w + 1$, and iterate Steps 2–4 until $w > W$.

Step 6: Set the objective values of π as:

$$\bar{A}(\pi) := \bar{A}(\pi)/W, \quad \bar{B}(\pi) := \bar{B}(\pi)/W.$$

In Step 3, we need to compute the objective values of a given sample. To do such work, we adopt a discrete event-based simulation (DES) method (Choi and Kang 2013). DES is actuated through a sequence of events. According to the parameterise event graph method proposed in the work of Choi and Kang (2013), we illustrate the process of DES regarding a system in Figure A1 in the Appendix. DES returns job completion time that is employed to straightforwardly calculate the makespan and total tardiness of a sample associated with a given solution.

Through using the above evaluation method, the objective values of solutions can be well evaluated.

4.3. Population initialisation

Population diversity plays important roles in multi-objective optimisation since an algorithm needs to search multiple optimal solutions in a solution space. To acquire a high-quality individual, this work adopts the earliest due date rule (Pinedo 2016) for producing the job sequence string where all jobs are sequenced in accordance with their due date from small to large, while the factory assignment string is randomly generated. The

other individuals are randomly generated as: Each element in the factory assignment string is generated from $[1, f]$ at random, and then a permutation having n job indexes is randomly generated as the job sequence string. Via using this method, Q individuals are produced as a population (recorded as $\mathbb{P}$ for convenience). In addition, an external archive (referred to as $\mathbb{E}$ for short) is developed to conserve obtained non-dominated individuals. At each iteration, we update $\mathbb{E}$ with all newly produced individuals via employing the Pareto rule (Deb et al. 2002).

4.4. Population-based search process

MOEA-EM performs a search process via a population to implement exploration and exploitation operations. It consists of an exploring search phase and an exploiting search phase of which detailed designs are given below.

4.4.1. Exploring search phase

At this phase, MOEA-EM explores promising regions from the entire search space. This work adopts a genetic operation including crossover and mutation to produce new individuals since it has a significant advantage in improving global search abilities (Fu et al. 2018). Generally, applying promising parents can increase the probability of generating better offspring individuals, and thus all individuals in $\mathbb{P}$ and $\mathbb{E}$ are allowed to be chosen as parent individuals. This work respectively names $\mathbb{P}$ and $\mathbb{E}$ as clusters 1 and 2, and we use the following method to choose parent individuals: (1) A binary selection approach (Fu et al. 2018) is employed to select a cluster with a smaller index; (2) If the selected cluster is $\mathbb{E}$, an individual is randomly chosen from it as one parent. Otherwise, we use the binary selection approach to choose a better individual from $\mathbb{P}$ as one parent. Using this method, we can select two parent individuals recorded as π_1 and π_2 . To generate an offspring individual π_o , two crossover approaches are respectively used to recombine factory assignment and job sequence strings. They are described as: (1) For the factory assignment string, a binary integer string is randomly generated and the factory indexes corresponding to 1 in π_1 are directly inherited into π_o , and the rest in π_o are chosen from π_2 . (2) For the job sequence string, an order-based crossover method (Fu et al. 2018) is employed. It randomly chooses two cut points, and the jobs between them in π_1 are directly inherited by π_o . Then, the remaining jobs are taken from π_2 as their relative sequence. An example is shown in Figure A2 in the Appendix.

After performing the crossover operation, MOEA-EM carries out a mutation operation on new individuals with probability p_m . This work designs four neighbourhood

structures to implement the mutation. For the factory assignment string, we use two methods as: (1) A job is selected, and we assign it to another factory at random; (2) Two positions are chosen at random, and the factory indexes on them are swapped. For the job sequence string, we employ two approaches as: (3) Two positions are chosen at random, and the job indexes on them are swapped; (4) Two positions are chosen at random, and the job index on a position is inserted to the other. By using the genetic operation, we can generate Q new individuals.

4.4.2. Exploiting search phase

In this phase, this research develops an iterated local search method (ILSM) to improve favourable individuals. It adopts four neighbourhood structures used in the mutation operation to find new individuals. Like the SA method, ILSM uses five conditions (Lin and Ying 2013), and a neighbourhood individual can substitute the present one on condition that any one of them is met. Let π_1 and π_2 be the current and neighbourhood individuals. The five conditions are given as:

-
- a. $\Delta O_1 \leq 0$ and $\Delta O_2 \leq 0$.
 - b. $\Delta O_1 > 0$, $\Delta O_2 \leq 0$ and $r < \exp(-\Delta O_1/L_e t_0)$.
 - c. $\Delta O_1 \leq 0$, $\Delta O_2 > 0$ and $r < \exp(-\Delta O_2/L_e t_0)$.
 - d. $\Delta O_1 > 0$, $\Delta O_2 > 0$, $\Delta O_1/O_1(\pi_2) < \Delta O_2/O_2(\pi_2)$ and $r < \exp(-\Delta O_2/L_e t_0)$.
 - e. $\Delta O_1 > 0$, $\Delta O_2 > 0$, $\Delta O_1/O_1(\pi_2) \geq \Delta O_2/O_2(\pi_2)$ and $r < \exp(-\Delta O_1/L_e t_0)$.
-

where $\Delta O_q = O_q(\pi_2) - O_q(\pi_1)$, $O_q(\pi_1)$ and $O_q(\pi_2)$ respectively denote the q -th objective values of π_1 and π_2 , $q \in \{1, 2\}$. L_e is the length of a solution equalling to $2 \cdot n$ for the studied problem. r is produced at random from an interval $[0, 1]$, and t_0 is a temperature coefficient generated from $[0.30, 0.75]$. Different from the SA approach, ILSM does not adopt the annealing method due to the restricted computing resources.

Algorithm 1: Exploiting search phase.

Input: $\mathbb{P}$ and $\mathbb{E}$.

Output: a new $\mathbb{E}$.

Begin

choose a non-dominated individual π from $\mathbb{P}$.

$\mathbb{N} := \emptyset$, $\pi_1 := \pi$.

for ($i := 0$ to l_M) **do**

$\pi_2 := \pi_1$.

create a number r from $[0, 1]$ at random.

produce a neighbour π_2 using one of four neighbourhood structures.

if (one of the five conditions is met) **then**

$\pi_1 := \pi_2$.

endif

$\mathbb{N} := \mathbb{N} \cup \{\pi_2\}$.

endfor

update $\mathbb{E}$ by using all individuals in $\mathbb{N}$.

End.

Algorithm 1 provides the main procedure of the exploiting search phase. Therein, I_M indicates the quantity of new produced individuals equalling to $2 \cdot \tau \cdot n$, and τ is a given parameter. All new-generated individuals at the exploring and exploiting stages are combined with the population, and we choose Q individuals for constructing the next population via adopting domination and crowding distance approaches (Deb et al. 2002).

4.5. External archive-based search

In MOEA-EM, $\mathbb{E}$ is employed to store all the non-dominated individuals acquired in the previous search process. Obviously, the individuals in $\mathbb{E}$ locate superior domains in the solution space, and it is necessary to search around them to find more better solutions. Thus, an external archive-based search (EABS) is performed to refine these individuals, aiming to further reinforce the exploitation performance of MOEA-EM. This work uses the ILSM to implement EABS. At each iteration, ILSM is used to refine $\kappa \cdot |\mathbb{E}|$ individuals selected from $\mathbb{E}$, where $|\mathbb{E}|$ denotes the quantity of individuals in $\mathbb{E}$, and κ is a given coefficient.

4.6. Procedure of MOEA-EM

Having introduced the designed MOEA and simulation methods in MOEA-EM, we summarise the entire procedure in Figure 3. Firstly, all parameters are initialised, and a group of individuals is generated as a population. Then, the evaluation approach is called to assess the objective values of individuals in the population. Secondly, MOEA-EM enters the following iterations: population-based and external archive-based search processes are performed in turn. The evaluation method needs to be called to assess the objective values of newly generated individuals. Finally, all individuals in the external archive are output if a pre-set termination criterion is met. Through the close cooperation of the designed MOEA and evaluation method, the considered DFJSP can be successfully solved, and all non-dominated solutions are provided for making decisions.

5. Experiments and discussions

5.1. Experimental setup

To examine the MOEA-EM in handling the DFJSP of interest, we conduct experiments via extending a group of benchmark problems. The prior studies do not deal with the problem under consideration, and a straightforward way to extend the existing benchmark test problems is needed. This work constructs a set of test instances

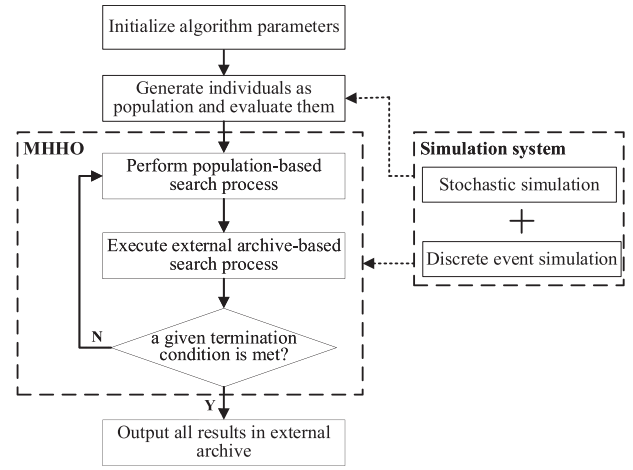


Figure 3. Procedure of the MOEA-EM.

in accordance with the benchmark series (la01-la40) introduced by Hurink, Jurisch, and Thole (1994). Following them, this work sets the average and maximum quantity of candidate machines to 3. The cases la01, la05, la10, la15, la20, la25, la30, la35, and la40 are selected and their job counts are duplicated as our test instances. Thus, the quantity of jobs belongs to 20, 30, 40, 60, the quantity of factories is from 2, 4, 6, and the quantity of machines at factories is from 5, 10, 15. Job processing time follows truncated normal distributions. This work takes the mean of job processing time as the same as the benchmark instances, while multiplying them and a coefficient equalling to 0.0001 is taken as their variance. To set job due date, we produce 20 solutions at random, and calculate the average job completion time across them. Then, the job due date equals to a product of their respective average completion time and a coefficient value of 2.0.

This work chooses four extensively noted MOEAs as rival approaches: NSGA-II (Deb et al. 2002), MOEA/D (Zhang and Li 2007), BMSA (Lin and Ying 2013) and IMA (Luo et al. 2022). NSGA-II is a proverbial MOEA via using domination strategies, while MOEA/D is an MOEA according to decomposition approaches. They have been widely adopted as benchmark methods for comparisons in solving a variety of complex optimisation problems (Fu et al. 2021b), and their excellent performance has been fully verified. Thus, we choose them as competitive methods. In addition, the ILSM in MOEA-EM uses five acceptance conditions derived from BMSA, and thereby BMSA is also selected as a peer method. Besides, IMA is a recently published work for solving DFJSPs, and it can be straightforwardly extended to deal with the considered problems. In consequence, we select IMA for comparisons as well. NSGA-II and MOEA/D use

the same solution representation, crossover and mutation methods with MOEA-EM, and BMSA also employs the same solution representation and generates neighbourhood solutions with the mutation methods. IMA uses the solution representation and decoding methods as its original work in Luo et al. (2022), and the corresponding genetic operations and local search methods are employed as well. All methods are coded with C++ on VC++ 2019 platform, and conducted using a computer having Intel Core i5-6500 CPU @ 3.20 GHz and 8GB RAM.

The computation complexity of MOEA-EM and its competitive approaches is analysed and reported in the Appendix. It is seen that MOEA-EM's computation complexity is not very great, and consequently it can be employed for settling the problem under consideration.

5.2. Performance metrics

Generally, multi-objective optimisation problems consist of many potentially optimal solutions. An MOEA should seek a group of non-dominated solutions having the features of more remarkable convergence and diversification to cover optimal solutions as much as possible. Thus, both the convergence and diversification of obtained non-dominated solutions need to be assessed. This work adopts IGD- (Deb et al. 2002) and Hypervolume-metric (Zhang and Li 2007) as evaluation indices. Notice that a smaller IGD implies better, while a larger Hypervolume means better. This article provides the calculation methods of two metrics in the Appendix.

5.3. Parameter setting

To investigate the effects of user parameters on MOEA-EM, this work adopts a Taguchi experiment method (Montgomery 2008) to ascertain the settings of pivotal parameters: Q , p_m , τ and κ . Each of them has four levels: $Q \in \{20, 40, 60, 80\}$, $p_m \in \{0.05, 0.10, 0.15, 0.20\}$, $\tau \in \{0.5, 1.0, 1.5, 2.0\}$ and $\kappa \in \{0.1, 0.4, 0.7, 1.0\}$. Hence, an $L_{16}(4^4)$ experiment is carried out, and Table A5 in the Appendix gives 16 parameter combinations. MOEA-EM using each combination solves an instance (namely la25 with 4 factories) 20 times. This article chooses the quantity of fitness evaluations as suspensive criteria set to $400 \cdot n$, in which n is the number of jobs. It indicates that more computation resources are expended as the number of jobs increases. The IGD is regarded as a response value (RV), while the average IGD across 20 times is treated as an average RV (ARV).

Tables A5 and A6 in the Appendix report the experimental results in regard to MOEA-EM. Besides, Figure 4 diagrams the influence of the four parameters

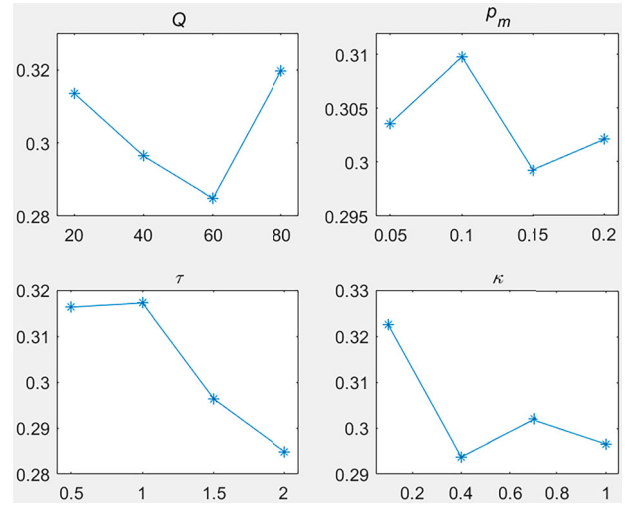


Figure 4. Influence trend of parameters.

on MOEA-EM. In accordance with the obtained outcome and dissection, we see that $Q = 60$, $p_m = 0.15$, $\tau = 2.0$ and $\kappa = 0.4$ are a better combination. They are used in the following experiments.

To make fair comparisons between MOEA-EM and its rival approaches, the DOE is also adopted to determine the rivals' promising parameters. Tables A7–A12 in the Appendix report the experimental results. Through dissecting the results, this work sets the rivals' parameters as: In regard to NSGA-II, we let its population size and mutation rate be severally equal to 60 and 0.10. With respect to MOEA/D, its population size, mutation rate and neighbourhood size are severally equal to 60, 0.15 and 20. Regarding BMSA, the quantity of solutions, initial temperature and annealing coefficient are respectively equal to 9, 7.5 and 0.90. As regards IMA, its population size, crossover probability, mutation probability and local search rate are set to 100, 0.8, 0.3 and 0.1, respectively.

5.4. Effectiveness of external archive-based search process

In MOEA-EM, an EABS is used to refine individuals in $\mathbb{E}$ to improve MOEA-EM's exploitation performance. To verify the role played by EABS, we conduct experiments to compare MOEA-EM with and without EABS (MOEA-EM-w/o-EABS). Table A13 in the Appendix reports the results regarding IGD and Hypervolume metrics.

It is observed that MOEA-EM beats MOEA-EM-w/o-EABS on solving all the instances. To further verify the effect of EABS in MOEA-EM, we execute the Friedman test and Nemenyi test (Brown and Mues 2012). Table 1 reports the statistical results. We can see that the MOEA-EM bests MOEA-EM-w/o-EABS regarding IGD- and Hypervolume-metric. Noteworthily, the difference is

Table 1. Results of Friedman and Nemenyi test via mean results for IGD- and Hypervolume-metric.

Algorithms	IGD-metric			Hypervolume-metric		
	Average rank value	Final rank	Difference	Average rank value	Final rank	Difference
MOEA-EM	1	1	1	1	1	1
MOEA-EM-w/o-EABS	2	2		2	2	

Table 2. Results of Wilcoxon signed rank test via mean results for IGD- and Hypervolume-metric.

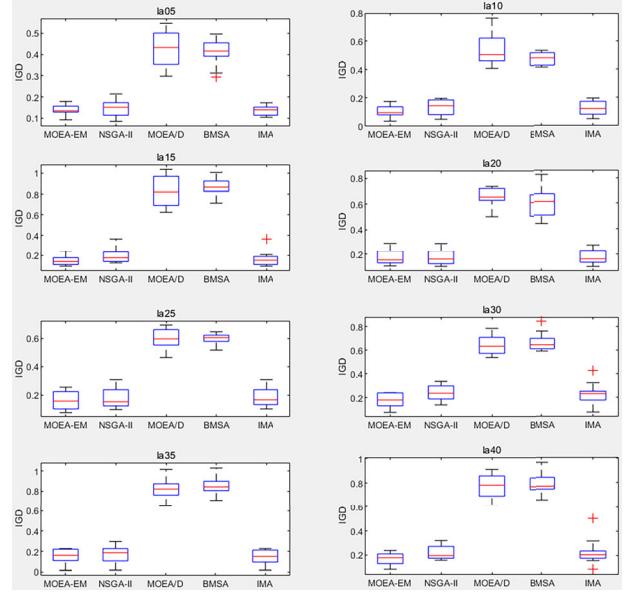
MOEA-EM vs. MOEA- EM-w/o-EABS	+ / ~ / -	R^+	R^-	Z	p-value	$\alpha = 0.05$
IGD-metric	27/0/0	378	0	4.54	0.0000	Yes
Hypervolume-metric	27/0/0	378	0	4.54	0.0000	Yes

equal to 1 that is larger than the critical value 0.3771, implying the significant difference. Furthermore, we employ the Wilcoxon signed rank test (Wang et al. 2020) to dissect their performance as provided in Table 2. The results in column ‘+ / ~ / -’ reveal that MOEA-EM outperforms MOEA-EM-w/o-EABS on 27 instances as regards IGD and Hypervolume metrics, respectively. Moreover, MOEA-EM acquires the bigger ‘ R^+ ’ value compared with the ‘ R^- ’ value, and the obtained p values between MOEA-EM and its peer are smaller than 0.05, implying that there exists the obvious gap between MOEA-EM and MOEA-EM-w/o-EABS. Hence, it can be claimed with confidence that MOEA-EM significantly bests MOEA-EM-w/o-EABS. Namely, the EABS plays very essential parts in MOEA-EM.

5.5. Experimental results and analysis

In the following, this work compares MOEA-EM with four existing algorithms: NSGA-II, MOEA/D, BMSA, and IMA. The outcome as regards IGD and Hypervolume metrics is respectively reported in Tables A14 and A15 in the Appendix. The results indicate that MOEA-EM can obtain well-converged and well-diversified solutions for the considered problem compared with its rivals. Figure 5 draws the boxplot diagrams of MOEA-EM and its competitors regarding IGD on the instances with six factories. It is seen that MOEA-EM is more superior and more stable than the rivals.

To further analyse the performance of MOEA-EM and its rivals with respect to IGD-metric, we use the Friedman test to obtain the average rank values and final rank values of all methods on solving all the instances. Table 3 reports the results. The average rank values of MOEA-EM and the rivals are severally 1.2963, 2.5185, 4.4074, 4.5926 and 2.2222. MOEA-EM ranks the first, and subsequent methods are IMA, NSGA-II, MOEA/D and BMSA. Thereby, it is declared that the capacity of

**Figure 5.** Boxplot graphs of five approaches for solving test problems with six factories regarding IGD-metric.**Table 3.** Friedman test results via mean values of IGD-metric.

Algorithms	IGD-metric	
	Average ranking value	Final rank
MOEA-EM	1.2963	1
NSGA-II	2.5185	3
MOEA/D	4.4074	4
BMSA	4.5926	5
IMA	2.2222	2

five approaches has significant difference. Furthermore, the Wilcoxon signed rank test is employed to dissect the results of MOEA-EM and its competitors as regards IGD-metric. Table 4 reports the outcome. The results in column ‘+ / ~ / -’ exhibit that MOEA-EM bests NSGA-II, MOEA/D, BMSA, and IMA on 23, 27, 27 and 26 out of 27 instances. Besides, the ‘ R^+ ’ values of MOEA-EM are far larger compared with the ‘ R^- ’ values, and the obtained p values between MOEA-EM and its peers are smaller than 0.05, indicating that the difference is obviously significant.

To analyse the capacity of MOEA-EM and its competitors regarding Hypervolume metric, we employ the Friedman test to obtain the average rank values and final rank values of all methods for solving the instances. Table 5 reports the results. MOEA-EM and its peers get

Table 4. Wilcoxon signed rank test results via mean values of IGD-metric.

MOEA-EM vs.	+ / ~ / -	R^+	R^-	Z	p-value	$\alpha = 0.05$
NSGA-II	23/0/4	334	44	3.48	0.0003	Yes
MOEA/D	27/0/0	378	0	4.54	0.0000	Yes
BMSA	27/0/0	378	0	4.54	0.0000	Yes
IMA	24/0/3	336	42	3.53	0.0004	Yes

Table 5. Friedman test results via mean values of Hypervolume-metric.

Algorithms	Hypervolume-metric	
	Average ranking value	Final rank
MOEA-EM	1.2222	1
NSGA-II	2.6296	3
MOEA/D	4.5741	5
BMSA	4.4259	4
IMA	2.1481	2

Table 6. Wilcoxon signed rank test results via mean values of Hypervolume-metric.

MOEA-EM vs.	+ / ~ / -	R^+	R^-	Z	p-value	$\alpha = 0.05$
NSGA-II	25/0/2	363	15	4.18	0.0000	Yes
MOEA/D	27/0/0	378	0	4.54	0.0000	Yes
BMSA	27/0/0	378	0	4.54	0.0000	Yes
IMA	23/0/4	332	46	3.44	0.0006	Yes

the average rank values 1.2222, 2.6296, 4.5741, 4.4259, and 2.1481, respectively. MOEA-EM also gets the first rank, followed by IMA, NSGA-II, BMSA, and MOEA/D. Furthermore, the Wilcoxon signed rank test is employed to dissect the outcome of MOEA-EM and its competitors regarding Hypervolume-metric. The results are provided in Table 6, in which the ' R^+ ', ' R^- ', and p values of all the pairwise comparisons regarding MOEA-EM are shown. As the table states, MOEA-EM significantly defeats NSGA-II, MOEA/D, BMSA, and IMA on 25, 27, 27, and 23 instances. Additionally, the statistical results reveal that the ' R^+ ' values of MOEA-EM are much larger than the ' R^- ' values for each pairwise comparison. Besides, the obtained p values show that MOEA-EM has a significant improvement over its peers with a level of significance $\alpha = 0.05$. To sum up, MOEA-EM is superior to its peers with respect to Hypervolume-metric.

In order to exhibit the experimental results clearly, we additionally use the Set Coverage, Spread and Generational Distance metrics (Ma et al. 2023b; Pan, Lei, and Wang 2020) to further analyse the acquired outcome. The results and analysis are provided in the Supplementary File (Fu et al. 2023c). It is noted that MOEA-EM also achieves better performance than its rivals regarding the three metrics in solving the considered problem. To better visualise the experimental results, Figure 6 shows distribution graphs of test instances with different

Table 7. Comparison results of MOEA-EM and CPLEX.

Instances	weighted vectors	CPLEX	MOEA-EM	
		AOV	AOV	Time (sec.)
5	(1.0, 0.0)	53.00*	70.00	28.32
	(0.5, 0.5)	31.00*	49.20	29.41
	(0.0, 1.0)	6.00*	10.00	27.21
7	(1.0, 0.0)	60.00	62.00	39.52
	(0.5, 0.5)	60.50	56.10	44.52
	(0.0, 1.0)	131.00	122.00	38.28
10	(1.0, 0.0)	–	95.00	67.32
	(0.5, 0.5)	–	90.50	74.33
	(0.0, 1.0)	–	232.00	65.67

factories. We detect that MOEA-EM can gain the non-dominated solutions with more excellent convergence and diversification than the rival methods.

5.6. Comparisons of MOEA-EM and CPLEX

To verify the ability of MOEA-EM in reaching optimal solutions for the considered problem, this work chooses a mathematical optimisation software, CPLEX, for comparisons. The problem under consideration has both multiple optimisation objectives and randomness. Thus, we transform it into three single-objective optimisation problems having deterministic job processing time via employing a weighted sum approach with three weight vectors: (1.0, 0.0), (0.5, 0.5) and (0.0, 1.0). This work chooses a case in a construction equipment manufacturing process for test. In this case, we must allocate 10 jobs to two factories for fabrication. Each factory has three machines, and each job owns three operations. In the Appendix, Table A16 provides the basic data regarding job processing time and due date. Using this case, we respectively construct three instances with the first 5, the first 7, and 10 jobs for each single-objective optimisation problem. Thus, we have 9 instances as given in Table 7.

According to the pilot studies, we can see that CPLEX cannot achieve the optimal solutions of the instance with 7 jobs after 3 h. Thus, we use the running time as CPLEX's termination condition equalling to 3 h and its maximum memory size is set to 1024M, whereupon CPLEX just reaches an approximated optimal value (AOV) for instances with more than 7 jobs. MOEA-EM addresses each instance 20 times independently, and all non-dominated solutions found across 20 times are collected. Then, the best solution associated with each weighted vector is chosen as the acquired solution of the corresponding instance. Besides, the running time of MOEA-EM in solving each instance over 20 times is recorded.

Table 7 reports the experimental results. The symbol '*' means that the corresponding approach can acquire an optimum for the instance, while '–' indicates that the

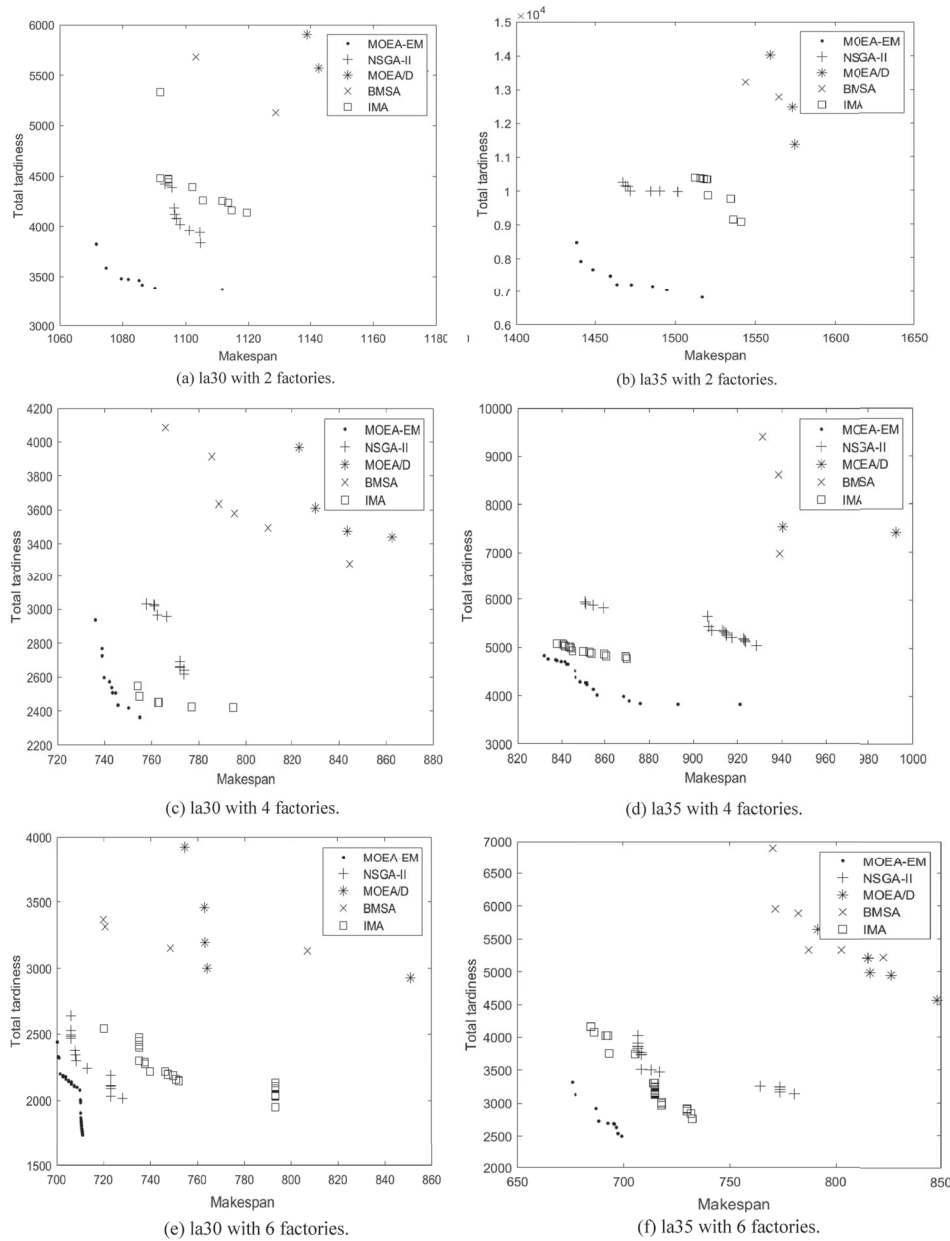


Figure 6. Distributed diagrams of non-dominated solutions acquired via four methods.

method cannot give a viable solution for the instance. Through observing the results, we see that CPLEX is able to achieve the optimum for the instances with 5 jobs, while it just reaches the approximated value for the instances with 7 jobs. For the instances with 10 jobs, CPLEX cannot find feasible results under the given conditions. MOEA-EM can obtain more excellent results than CPLEX for the instances with more than 7 jobs except for the instance associated with the weighted vector (1.0, 0.0). Notice that MOEA-EM is slightly worse than CPLEX for the instances with 5 jobs. By dissecting the results, it can be inferred that MOEA-EM performs more superiorly in solving the studied problem with more than 7 jobs via using less running time. Hence,

the validation results of MOEA-EM in solving industrial cases can gain acceptance.

In accordance with the afore-mentioned results and dissection, we can declare that MOEA-EM can beat its competitive approaches, and thus it is a much more excellent solver for tackling the problem under consideration. MOEA-EM adopts population-based and external archive-based search processes to search a solution space. The former explores the entire solution space to find promising regions, while the latter focuses on exploiting the found promising regions. In addition, MOEA-EM uses an evaluation method including stochastic simulation and discrete event simulation to assess obtained solutions. Analysis of the experimental results suggests

that MOEA-EM can successfully and effectively deal with our considered problem via close cooperation of the above methods.

6. Conclusions

This work considers scheduling a distributed flexible job shop having random processing time to reach minimal makespan and total tardiness. A stochastic programming model is established to define the problem. To work out it, a multi-objective evolutionary algorithm incorporating an evaluation method is devised. The competitiveness of the designed method is fully validated via performing simulation experiment and comparison analysis. The developed model and approach can give assistance to managers and decision-makers for making more informed and more reliable solutions in real-life environments.

Future work will seek to (1) formulate DFJSPs with both stochasticity and dynamic disruptions for improving processing quality and production resilience; (2) develop effective search algorithms and simulation approaches to settle the investigated problem; and (3) design smart contracts and resource scheduling methods for manufacturing enterprises using blockchain technology to form collaborative production networks.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available from the corresponding author, [Gao, K. Z.], upon reasonable request.

References

- Azab, A., and B. Naderi. 2014. "Greedy Heuristics for Distributed Job Shop Problems." *Procedia CIRP* 20: 7–12. <https://doi.org/10.1016/j.procir.2014.05.025>
- Badakhshan, E., and P. Ball. 2023. "Deploying Hybrid Modelling to Support the Development of a Digital Twin for Supply Chain Master Planning Under Disruptions." *International Journal of Production Research* 62 (10): 3606–3637. <https://doi.org/10.1080/00207543.2023.2244604>
- Becerra, P., J. Mula, and R. Sanchis. 2023. "Optimising Location, Inventory and Transportation in a Sustainable Closed-Loop Supply Chain." *International Journal of Production Research* 62 (5): 1609–1632. <https://doi.org/10.1080/00207543.2023.2197515>
- Brown, I., and C. Mues. 2012. "An Experimental Comparison of Classification Algorithms for Imbalanced Credit Scoring Data Sets." *Expert Systems with Applications* 39 (3): 3446–3453. <https://doi.org/10.1016/j.eswa.2011.09.033>
- Carlucci, D., P. Renna, and S. Materi. 2023. "A Job-Shop Scheduling Decision-Making Model for Sustainable Production Planning with Power Constraint." *IEEE Transactions on Engineering Management* 70 (5): 1923–1932. <https://doi.org/10.1109/TEM.2021.3103108>
- Chang, H. C., and T. K. Liu. 2017. "Optimisation of Distributed Manufacturing Flexible Job Shop Scheduling by Using Hybrid Genetic Algorithms." *Journal of Intelligent Manufacturing* 28 (8): 1973–1986. <https://doi.org/10.1007/s10845-015-1084-y>
- Chaouch, I., O. B. Driss, and K. Ghedira. 2017. "A Modified Ant Colony Optimization Algorithm for the Distributed Job Shop Scheduling Problem." *Procedia Computer Science* 112: 296–305. <https://doi.org/10.1016/j.procs.2017.08.267>
- Chaouch, I., O. B. Driss, and A. Ghedira. 2019. "A Novel Dynamic Assignment Rule for the Distributed Job Shop Scheduling Problem Using a Hybrid Ant-Based Algorithm." *Applied Intelligence* 49 (5): 1903–1924. <https://doi.org/10.1007/s10489-018-1343-7>
- Choi, T. M., A. Dolgui, D. Ivanov, and E. Pesch. 2022. "OR and Analytics for Digital, Resilient, and Sustainable Manufacturing 4.0." *Annals of Operations Research* 310 (1): 1–6. <https://doi.org/10.1007/s10479-022-04536-3>
- Choi, B. K., and D. Kang. 2013. *Modeling and Simulation of Discrete Event Systems*. South Korea: John Wiley & Sons.
- Deb, K., A. Pratap, S. Agarwal, and T. Meyarivan. 2002. "A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II." *IEEE Transactions on Evolutionary Computation* 6 (2): 182–197. <https://doi.org/10.1109/4235.996017>
- De Giovanni, L., and F. Pezzella. 2010. "An Improved Genetic Algorithm for the Distributed and Flexible Job-Shop Scheduling Problem." *European Journal of Operations Research* 200 (2): 395–408. <https://doi.org/10.1016/j.ejor.2009.01.008>
- Du, Y., J. Q. Li, C. Luo, and L. L. Meng. 2021. "A Hybrid Estimation of Distribution Algorithm for Distributed Flexible Job Shop Scheduling with Crane Transportations." *Swarm and Evolutionary Computation* 62. <https://doi.org/10.1016/j.swevo.2021.100861>
- Fu, Y. P., J. L. Ding, H. F. Wang, and J. W. Wang. 2018. "Two-Objective Stochastic Flow-Shop Scheduling with Deteriorating and Learning Effect in Industry 4.0-Based Manufacturing System." *Applied Soft Computing* 68: 847–855. <https://doi.org/10.1016/j.asoc.2017.12.009>
- Fu, Y. P., K. Z. Gao, L. Wang, M. Huang, Y. C. Liang, and H. Y. Dong. 2023c. Supplementary File of this Paper Can Be Downloaded from https://www.researchgate.net/publication/374506957_FYP-SF-IJPR-Oct_07_2023.
- Fu, Y. P., Y. S. Hou, Z. F. Wang, X. W. Xu, K. Z. Gao, and L. Wang. 2021a. "Distributed Scheduling Problems in Intelligent Manufacturing Systems." *Tsinghua Science and Technology* 26 (5): 625–645. <https://doi.org/10.26599/TST.2021.9010009>
- Fu, Y. P., H. B. Li, M. Huang, and H. Xiao. 2023a. "Bi-Objective Modeling and Optimization for Stochastic Two-Stage Open Shop Scheduling Problems in the Sharing Economy." *IEEE Transactions on Engineering Management* 70 (10): 3395–3409. <https://doi.org/10.1109/TEM.2021.3095954>
- Fu, Y. P., X. M. Ma, K. Z. Gao, Z. W. Li, and H. Y. Dong. 2023b. "Multi-Objective Home Health Care Routing and Scheduling with Sharing Service via a Problem-Specific Knowledge Based Artificial Bee Colony Algorithm." *IEEE Transactions on Intelligent Transportation Systems* 25 (2): 1706–1719. <https://doi.org/10.1109/TITS.2023.3315785>
- Fu, Y. P., G. D. Tian, G. A. M. Fathollahi-Fard, A. Ahmadi, and C. Y. Zhang. 2019. "Stochastic Multi-Objective Modelling and

- Optimization of an Energy-Conscious Distributed Permutation Flow Shop Scheduling Problem with the Total Tardiness Constraint." *Journal of Cleaner Production* 226: 515–525. <https://doi.org/10.1016/j.jclepro.2019.04.046>
- Fu, Y. P., M. C. Zhou, X. W. Guo, and L. Qi. 2020. "Scheduling Dual-Objective Stochastic Hybrid Flow Shop with Deteriorating Jobs via Bi-Population Evolutionary Algorithm." *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 50 (12): 5037–5048. <https://doi.org/10.1109/TSMC.2019.2907575>
- Fu, Y. P., M. C. Zhou, X. W. Guo, L. Qi, and K. Sedraoui. 2021b. "Multiverse Optimization Algorithm for Stochastic Biobjective Disassembly Sequence Planning Subject to Operation Failures." *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 52 (2): 1041–1051. <https://doi.org/10.1109/TSMC.2021.3049323>
- Geng, Y. D., and J. Q. Li. 2023. "A Knowledge-Driven Multiobjective Algorithm for Distributed Hybrid Flowshop with Group and Carryover Setup in Glass Manufacturing Systems." *Computers & Industrial Engineering* 181. <https://doi.org/10.1016/j.cie.2023.109325>
- Hou, Y. S., H. F. Wang, Y. P. Fu, K. Z. Gao, and H. Zhang. 2023. "Multi-Objective Brain Storm Optimization for Integrated Scheduling of Distributed Flow Shop and Distribution with Maximal Processing Quality and Minimal Total Weighted Earliness and Tardiness." *Computers & Industrial Engineering* 179. <https://doi.org/10.1016/j.cie.2023.109217>
- Hsu, C. Y., B. R. Kao, and K. R. Lai. 2016. "Agent-Based Fuzzy Constraint-Directed Negotiation Mechanism for Distributed Job Shop Scheduling." *Engineering Applications of Artificial Intelligence* 53: 140–154. <https://doi.org/10.1016/j.engappai.2016.04.005>
- Huang, Y. Y., Q. K. Pan, and L. Gao. 2023. "An Effective Memetic Algorithm for the Distributed Flowshop Scheduling Problem with an Assemble Machine." *International Journal of Production Research* 61 (6): 1755–1770. <https://doi.org/10.1080/00207543.2022.2047238>
- Hurink, J., B. Jurisch, and M. Thole. 1994. "Tabu Search for the Job-Shop Scheduling Problem with Multi-Purpose Machines." *Operations-Research-Spektrum* 15 (4): 205–215. <https://doi.org/10.1007/BF01719451>
- Ip, W. H., M. Huang, K. L. Yung, and D. W. Wang. 2003. "Genetic Algorithm Solution for a Risk-Based Partner Selection Problem in a Virtual Enterprise." *Computers & Operations Research* 30 (2): 213–231. [https://doi.org/10.1016/S0305-0548\(01\)00092-2](https://doi.org/10.1016/S0305-0548(01)00092-2)
- Jeong, I. J., and S. B. Yim. 2009. "A Job Shop Distributed Scheduling Based on Lagrangian Relaxation to Minimise Total Completion Time." *International Journal of Production Research* 47 (24): 6783–6805. <https://doi.org/10.1080/00207540701824217>
- Jiang, E. D., L. Wang, and Z. P. Peng. 2020. "Solving Energy-Efficient Distributed Job Shop Scheduling via Multi-Objective Evolutionary Algorithm with Decomposition." *Swarm and Evolutionary Computation* 58. <https://doi.org/10.1016/j.swevo.2020.100745>
- Lei, D. M., and B. Su. 2023. "A Multi-Class Teaching-Learning-Based Optimization for Multi-Objective Distributed Hybrid Flow Shop Scheduling." *Knowledge-Based Systems* 263. <https://doi.org/10.1016/j.knsys.2023.110252>
- Li, R., W. Y. Gong, L. Wang, C. Lu, and S. N. Jiang. 2022a. "Two-Stage Knowledge-Driven Evolutionary Algorithm for Distributed Green Flexible Job Shop Scheduling with Type-2 Fuzzy Processing Time." *Swarm and Evolutionary Computation* 74. <https://doi.org/10.1016/j.swevo.2022.101139>
- Li, X. Y., J. Xie, Q. J. Ma, L. Gao, and P. G. Li. 2022b. "Improved Gray Wolf Optimizer for Distributed Flexible Job Shop Scheduling Problem." *Science China Technological Sciences* 65 (9): 2105–2115. <https://doi.org/10.1007/s11431-022-2096-6>
- Lin, C. S., P. Y. Li, J. M. Wei, and M. C. Wu. 2020. "Integration of Process Planning and Scheduling for Distributed Flexible Job Shops." *Computers & Operations Research* 124. <https://doi.org/10.1016/j.cor.2020.105053>
- Lin, S. W., and K. C. Ying. 2013. "Minimizing Makespan and Total Flowtime in Permutation Flowshops by a Bi-Objective Multi-Start Simulated-Annealing Algorithm." *Computers & Operations Research* 40 (6): 1625–1647. <https://doi.org/10.1016/j.cor.2011.08.009>
- Liu, T. K., Y. P. Chen, and J. H. Chou. 2014. "Solving Distributed and Flexible Job-Shop Scheduling Problems for a Real-World Fastener Manufacturer." *IEEE Access* 2: 1598–1606. <https://doi.org/10.1109/ACCESS.2015.2388486>
- Lu, P. H., M. C. Wu, H. Tan, Y. H. Peng, and C. F. Chen. 2018. "A Genetic Algorithm Embedded with a Concise Chromosome Representation for Distributed and Flexible Job-Shop Scheduling Problems." *Journal of Intelligent Manufacturing* 29 (1): 19–34. <https://doi.org/10.1007/s10845-015-1083-z>
- Luo, Q., Q. W. Deng, G. L. Gong, X. Guo, and X. H. Liu. 2022. "A Distributed Flexible Job Shop Scheduling Problem Considering Worker Arrangement Using an Improved Memetic Algorithm." *Expert Systems with Applications* 207. <https://doi.org/10.1016/j.eswa.2022.117984>
- Luo, Q., Q. W. Deng, G. L. Gong, L. Zhang, W. W. Han, and K. X. Li. 2020. "An Efficient Memetic Algorithm for Distributed Flexible Job Shop Scheduling Problem with Transfers." *Expert Systems with Applications* 160. <https://doi.org/10.1016/j.eswa.2020.113721>
- Ma, X. M., Y. P. Fu, K. Z. Gao, H. Zhang, and J. H. Mou. 2023b. "A Knowledge-Based Multi-Objective Evolutionary Algorithm for Solving Home Health Care Routing and Scheduling Problems with Multiple Centers." *Applied Soft Computing* 144. <https://doi.org/10.1016/j.asoc.2023.110491>
- Ma, X. M., Y. P. Fu, K. Z. Gao, L. H. Zhu, and A. Sadollah. 2023a. "A Multi-Objective Scheduling and Routing Problem for Home Health Care Services via Brain Storm Optimization." *Complex System Modeling and Simulation* 3 (1): 32–46. <https://doi.org/10.23919/CSMS.2022.0025>
- Mahmoodjanloo, M., R. Tavakkoli-Moghaddama, A. Baboli, and A. Bozorgi-Amiri. 2022. "Distributed Job-Shop Rescheduling Problem Considering Reconfigurability of Machines: A Self-Adaptive Hybrid Equilibrium Optimizer." *International Journal of Production Research* 60 (16): 4973–4994. <https://doi.org/10.1080/00207543.2021.1946193>
- Marzouki, B., O. B. Driss, and K. Ghédira. 2018. "Solving Distributed and Flexible Job Shop Scheduling Problem Using a Chemical Reaction Optimization Metaheuristic." *Procedia Computer Science* 126: 1424–1433. <https://doi.org/10.1016/j.procs.2018.08.114>
- Meng, L. L., C. Y. Zhang, Y. P. Ren, B. Zhang, and C. Lu. 2020. "Mixed-Integer Linear Programming and Constraint

- Programming Formulations for Solving Distributed Flexible Job Shop Scheduling Problem.” *Computers & Industrial Engineering* 142. <https://doi.org/10.1016/j.cie.2020.106347>
- Montgomery, D. C. 2008. *Design and Analysis of Experiments*. 7th ed. Hoboken, NJ: Wiley.
- Naderi, B., and A. Azab. 2015. “An Improved Model and Novel Simulated Annealing for Distributed Job Shop Problems.” *The International Journal of Advanced Manufacturing Technology* 81 (1-4): 693–703. <https://doi.org/10.1007/s00170-015-7080-8>
- Neumann, A., A. Hajji, M. Rekik, and R. Pellerin. 2023. “Genetic Algorithms for Planning and Scheduling Engineer-to-Order Production: A Systematic Review.” *International Journal of Production Research* 62 (8): 2888–2817. <https://doi.org/10.1080/00207543.2023.2237122>
- Pan, Y. Y., K. Z. Gao, Z. W. Li, and N. Q. Wu. 2023. “Solving Biobjective Distributed Flow-Shop Scheduling Problems with Lot-Streaming Using an Improved Jaya Algorithm.” *IEEE Transactions on Cybernetics* 53 (6): 3818–3828. <https://doi.org/10.1109/TCYB.2022.3164165>
- Pan, Z. X., D. M. Lei, and L. Wang. 2020. “A Knowledge-Based Two-Population Optimization Algorithm for Distributed Energy-Efficient Parallel Machines Scheduling.” *IEEE Transactions on Cybernetics* 52 (6): 5051–5063. <https://doi.org/10.1109/TCYB.2020.3026571>
- Pan, Z. X., D. M. Lei, and L. Wang. 2022. “A Bi-Population Evolutionary Algorithm with Feedback for Energy-Efficient Fuzzy Flexible Job Shop Scheduling.” *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 52 (8): 5295–5307. <https://doi.org/10.1109/TSMC.2021.3120702>
- Pinedo, M. L. 2016. *Scheduling: Theory, Algorithms, and Systems*, fifth version. Cham: Springer International Publishing.
- Şahman, M. A. 2021. “A Discrete Spotted Hyena Optimizer for Solving Distributed Job Shop Scheduling Problems.” *Applied Soft Computing* 106. <https://doi.org/10.1016/j.asoc.2021.107349>
- Sang, Y. W., and J. P. Tan. 2022. “Intelligent Factory Many-Objective Distributed Flexible Job Shop Collaborative Scheduling Method.” *Computers & Industrial Engineering* 164. <https://doi.org/10.1016/j.cie.2021.107884>
- Shao, W. S., Z. S. Shao, and D. C. Pi. 2021. “Effective Constructive Heuristics for Distributed No-Wait Flexible Flow Shop Scheduling Problem.” *Computers & Operations Research* 136. <https://doi.org/10.1016/j.cor.2021.105482>
- Shi, W. Y., and C. Mena. 2023. “Supply Chain Resilience Assessment with Financial Considerations: A Bayesian Network-Based Method.” *IEEE Transactions on Engineering Management* 70 (6): 2241–2256. <https://doi.org/10.1109/TEM.2021.3066600>
- Siqin, T., T. M. Choi, S. H. Chung, and X. Wen. 2022. “Platform Operations in the Industry 4.0 Era: Recent Advances and the 3As Framework.” *IEEE Transactions on Engineering Management* 71: 1145–1162. <https://doi.org/10.1109/TEM.2021.3138745>
- Tang, H. T., B. Fang, R. Liu, Y. B. Li, and S. S. Guo. 2022. “A Hybrid Teaching and Learning-Based Optimization Algorithm for Distributed Sand Casting Job Shop Scheduling Problem.” *Applied Soft Computing* 120. <https://doi.org/10.1016/j.asoc.2022.108694>
- Thenarasu, M., K. Rameshkumar, M. Di Mascolo, and S. P. Anbuudayasankar. 2023. “Multi-criteria Scheduling of Realistic Flexible Job Shop: A Novel Approach for Integrating Simulation Modelling and Multi-Criteria Decision Making.” *International Journal of Production Research* 62 (1-2): 336–358. <https://doi.org/10.1080/00207543.2023.2238084>
- Tremblet, D., S. Thevenin, and A. Dolgui. 2023. “Makespan Estimation in a Flexible Job-Shop Scheduling Environment Using Machine Learning.” *International Journal of Production Research* 62 (10): 3654–3670. <https://doi.org/10.1080/00207543.2023.2245918>
- Wang, Y. H., Y. Y. Han, Y. T. Wang, M. F. Tasgetiren, J. Q. Li, and K. Z. Gao. 2023. “Intelligent Optimization Under the Makespan Constraint: Rapid Evaluation Mechanisms Based on the Critical Machine for the Distributed Flowshop Group Scheduling Problem.” *European Journal of Operational Research* 311 (3): 816–832. <https://doi.org/10.1016/j.ejor.2023.05.010>
- Wang, J., Y. Liu, S. Ren, C. Wang, and W. B. Wang. 2021b. “Evolutionary Game Based Real-Time Scheduling for Energy-Efficient Distributed and Flexible job Shop.” *Journal of Cleaner Production* 293. <https://doi.org/10.1016/j.jclepro.2021.126093>
- Wang, J. H., Y. Y. Sun, Z. Z. Zhang, and S. C. Gao. 2020. “Solving Multitrip Pickup and Delivery Problem with Time Windows and Manpower Planning Using Multiobjective Algorithms.” *IEEE/CAA Journal of Automatica Sinica* 7 (4): 1134–1153. <https://doi.org/10.1109/JAS.2020.1003204>
- Wang, J. J., and L. Wang. 2021a. “A Cooperative Memetic Algorithm with Learning-Based Agent for Energy-Aware Distributed Hybrid Flow-Shop Scheduling.” *IEEE Transactions on Evolutionary Computation* 26 (3): 461–475. <https://doi.org/10.1109/TEVC.2021.3106168>
- Wu, X. L., X. J. Liu, and N. Zhao. 2019. “An Improved Differential Evolution Algorithm for Solving a Distributed Assembly Flexible Job Shop Scheduling Problem.” *Memetic Computing* 11 (4): 335–355. <https://doi.org/10.1007/s12293-018-00278-7>
- Wu, Z. Y., R. Zhou, M. Goh, Y. Wang, Z. T. Xu, and W. Y. Song. 2023. “A Digital Twin-Based Modularized Design Approach for Smart Warehouses.” *International Journal of Computer Integrated Manufacturing*. <https://doi.org/10.1080/0951192X.2023.2278100>
- Xie, J., L. Gao, Q. K. Pan, and M. F. Tasgetiren. 2019. “An Effective Multi-Objective Artificial bee Colony Algorithm for Energy Efficient Distributed Job Shop Scheduling.” *Procedia Manufacturing* 39: 1194–1203. <https://doi.org/10.1016/j.promfg.2020.01.350>
- Xu, W. X., Y. W. Hu, W. Luo, L. Wang, and R. Wu. 2021. “A Multi-Objective Scheduling Method for Distributed and Flexible Job Shop Based on Hybrid Genetic Algorithm and Tabu Search Considering Operation Outsourcing and Carbon Emission.” *Computers & Industrial Engineering* 157. <https://doi.org/10.1016/j.cie.2021.107318>
- Xu, X. Y., S. P. Sethi, S. H. Chung, and T. M. Choi. 2022. “Reforming Global Supply Chain Management Under Pandemics: The GREAT-3Rs Framework.” *Production and Operations Management* 32 (2): 524–546. <https://doi.org/10.1111/poms.13885>
- Zhang, Q. F., and H. Li. 2007. “MOEA/D: A Multiobjective Evolutionary Algorithm Based on Decomposition.” *IEEE*

- Transactions on Evolutionary Computation* 11 (6): 712–731. <https://doi.org/10.1109/TEVC.2007.892759>
- Zhao, F. Q., R. Ma, and L. Wang. 2022. “A Self-Learning Discrete Jaya Algorithm for Multiobjective Energy-Efficient Distributed No-Idle Flow-Shop Scheduling Problem in Heterogeneous Factory System.” *IEEE Transactions on Cybernetics* 52 (12): 12675–12686. <https://doi.org/10.1109/TCYB.2021.3086181>
- Zhao, F. Q., Z. S. Xu, H. Z. Bao, T. P. Xu, N. N. Zhu, and Jonrinaldi. 2023. “A Cooperative Whale Optimization Algorithm for Energy-Efficient Scheduling of the Distributed Blocking Flow-Shop with Sequence-Dependent Setup Time.” *Computers & Industrial Engineering* 178. <https://doi.org/10.1016/j.cie.2023.109082>
- Zheng, J., L. Wang, and J. J. Wang. 2020. “A Cooperative Coevolution Algorithm for Multi-Objective Fuzzy Distributed Hybrid Flow Shop.” *Knowledge-Based Systems* 194. <https://doi.org/10.1016/j.knosys.2020.105536>